Overview on LoRa/LoRaWAN

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Internet of Things, Communication Engineering's Course

Automation is not what you think about...

- Software development in **C** / **C++** for electrical automation systems
- Linux kernel development and cybersecurity work
- Focus on critical infrastructure and energy networks

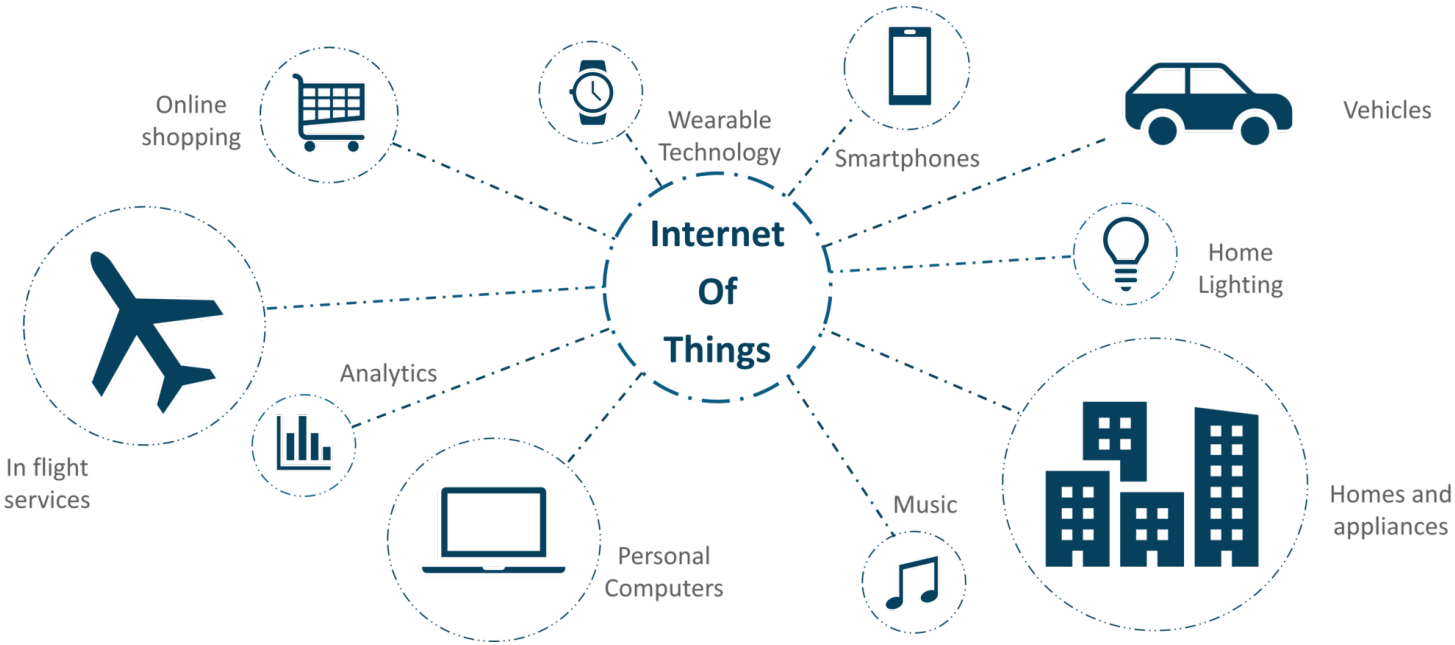
- Systems deployed for:
  - **Terna, Enel** (Italy)
  - **Enedis, RTE** (France)
  - **IPTO (i.e., ADMIE)** (Greece)

- Domain:
  - substation automation
  - protection and control systems
  - real-time industrial communication

Agenda

1. Low Power Wide Area Networks (LPWANs)
2. Use cases
3. LoRa
4. LoRaWAN
5. Current and Future development

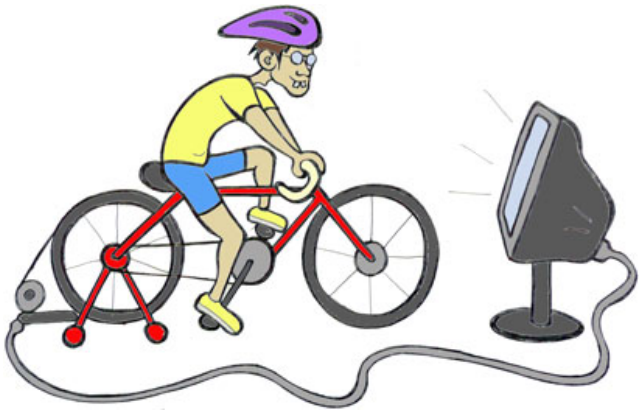
IoT applications



IoT $\rightarrow$ Connecting embedded devices

Challenges at the edge

Battery



Good range



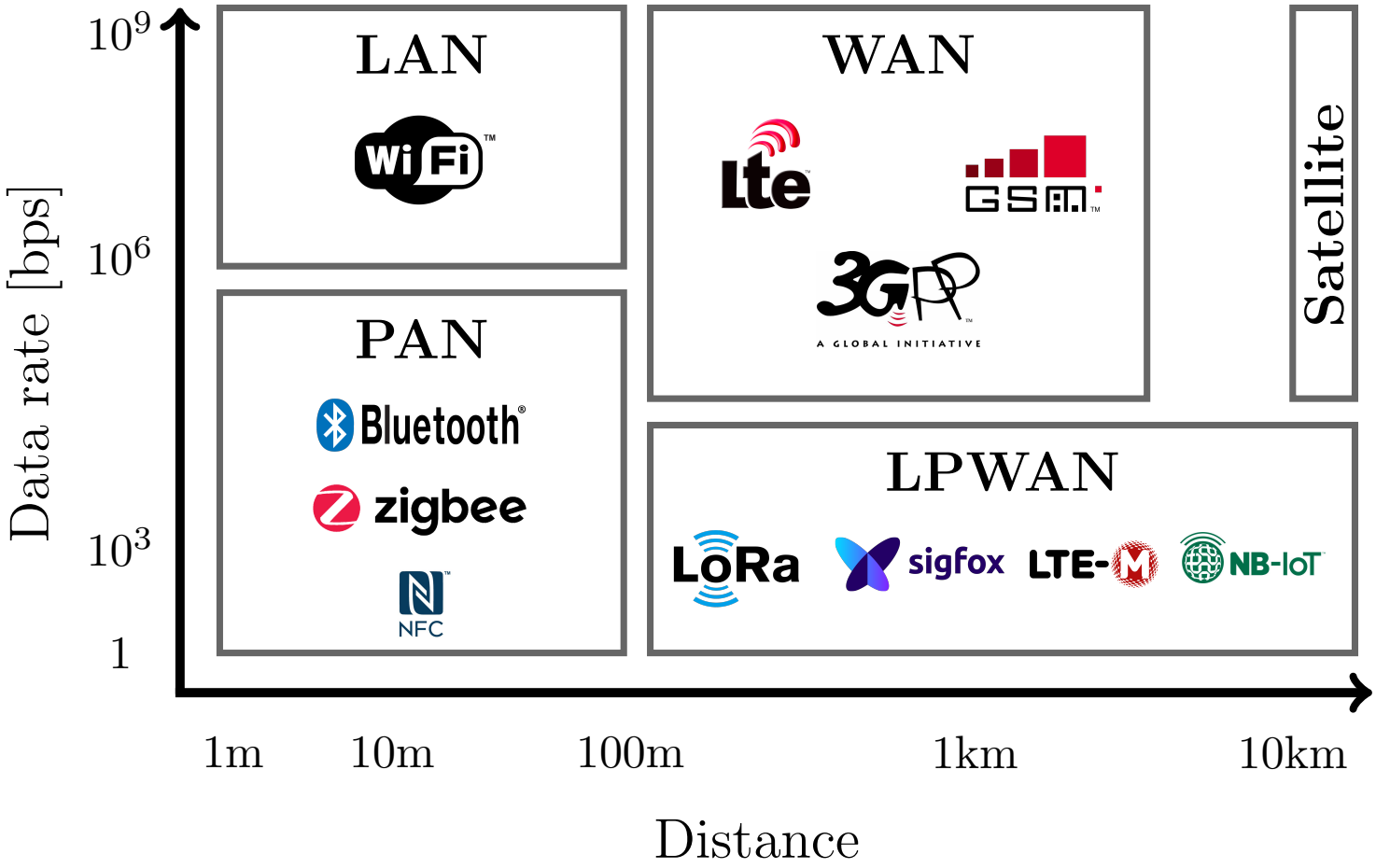
Security



Cost



Current technologies



Cellular technologies

Cellular networks are widely used for IoT (e.g. home alarm systems)

- Power consumption can be high
- Private deployments require spectrum/licenses

What about 2G?

Ubiquitous, low-cost, energy-aware $\rightarrow$ historically good fit for IoT

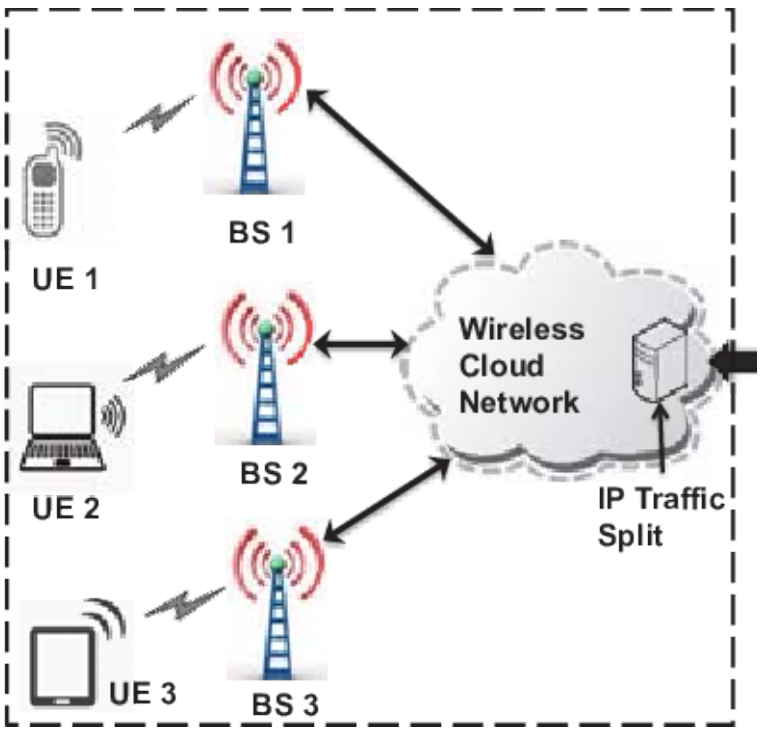
- But legacy networks are disappearing:
  - 2G: already shut down in several countries (e.g. US, Australia)
  - 3G: shut down in Italy

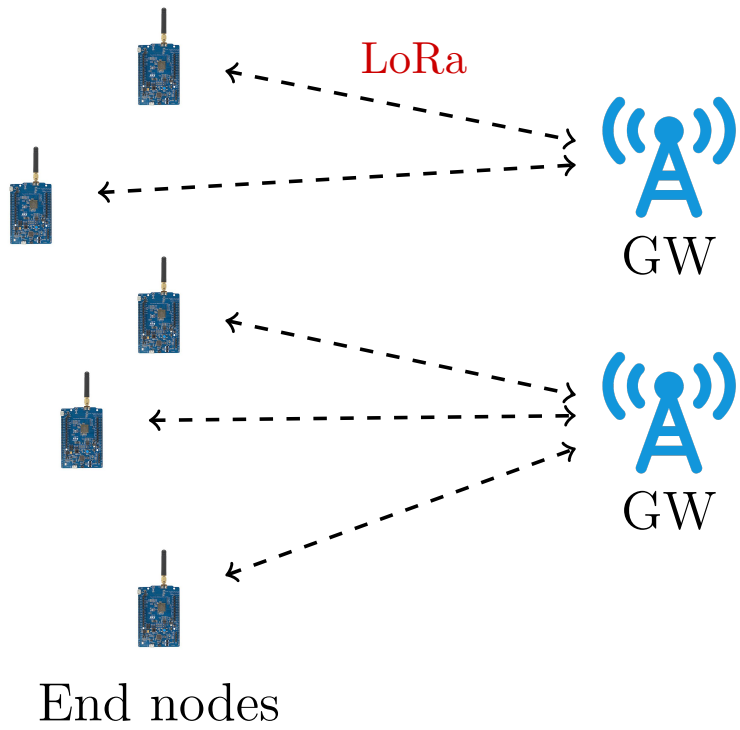
Legacy cellular is not a reliable IoT baseline

LPWAN

LPWAN: like cellular, but optimized for IoT/M2M (node is not connected to a BTS)

- network of GW/BTS
- star topology
- range in kilometers





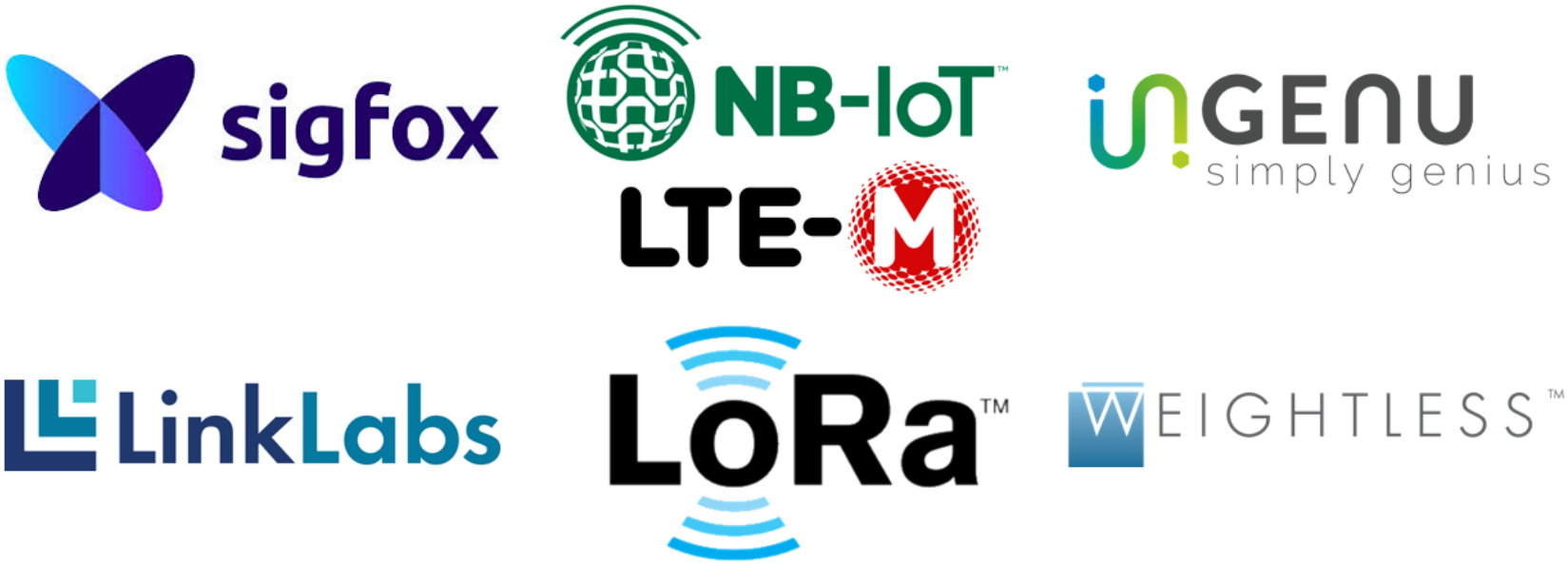
Main features of LPWANs

Key properties

- Low cost
- Low data rate
- Unlicensed spectrum $\rightarrow$ easy deployment

	Range	Battery lifetime
2G	$2 \div 4$ km	days
IEEE 802.15.4	$10 \div 100$ m	months–years
WiFi	$10 \div 50$ m	days
LPWAN	$10 \div 100$ km	years

LPWAN: emerging technologies



Different business models, technical specs (e.g., frequencies)

LoRaWAN recognized as standard for LPWAN by ITU

Definitions

LoRa: PHY layer

LoRaWAN: MAC layer

LoRa Alliance: defines LoRaWAN specifications

Technical comparison

	SigFox	LoRaWAN	NB-IoT
Modulation	BPSK	CSS	QPSK
Frequency	ISM	ISM	LTE bands
Max rate [kbps]	0.1	50	200
Max msg/day	140 (U), 4 (D)	Unlimited	Unlimited
Range [km]	$10 \div 40$	$5 \div 20$	$1 \div 10$
Handover	not applicable	not applicable	no
Localization	Yes (RSSI)	Yes (TDOA)	WIP
Allow private network	No	Yes	No

LPWAN Cost comparison

	SigFox	LoRaWAN	NB-IoT
Spectrum cost [M€/MHz]	free	free	500
Deployment cost [€/BTS]	> 4000	> 100	> 15000
End-device cost [€]	< 2	$3 \div 5$	> 20

LoRa*WAN* vs SigFox: business models

SigFox

- Fully vertically integrated company (end-to-end control)
- Closed ecosystem
- Network and cloud sold as a service
- Hardware ecosystem licensed under strict terms

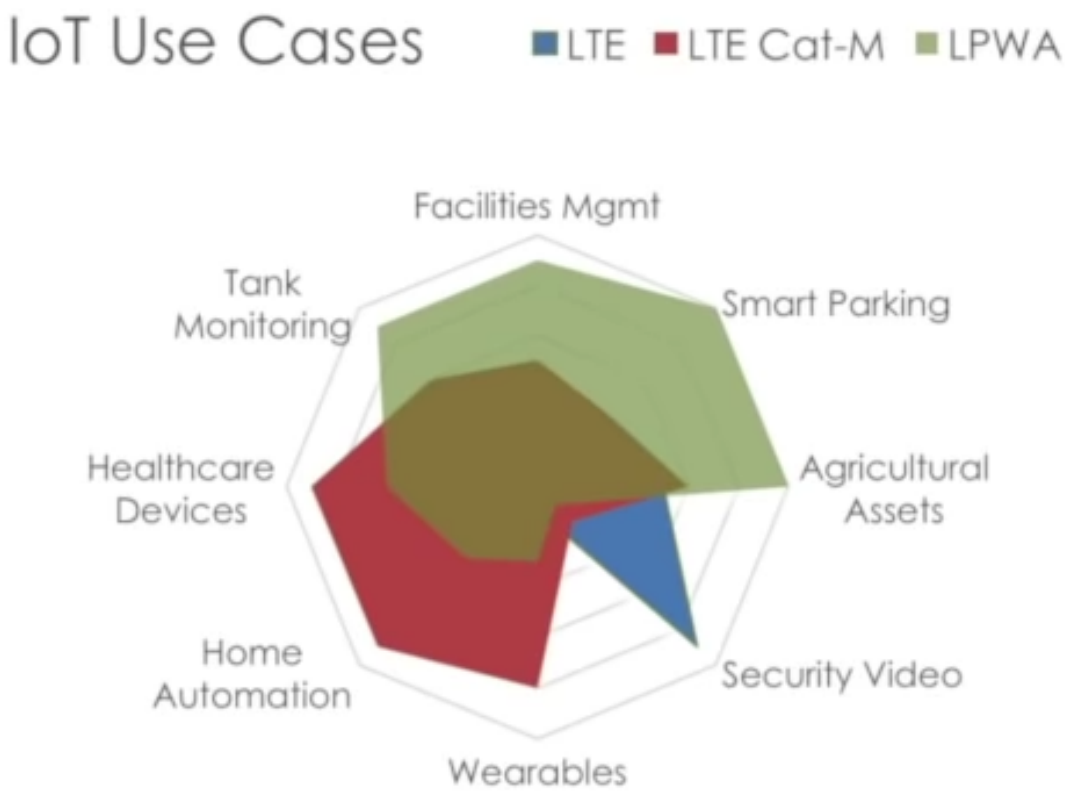
LoRaWAN

- Open specification defined by the LoRa Alliance
- Anyone can deploy networks or build compatible devices
- Semtech provides the radio chips, but ecosystem is multi-vendor
- Strong adoption also in research and private deployments

Key difference: control vs openness of the ecosystem

IoT use cases

- Smart city
  - Metering
  - Parking
  - Alarms
- Smart agriculture
  - Environmental sensors



Remark

Not only for range of km, think also about indoor with obstacles or underwater/underground (farm monitoring)

IoT use cases: smart farming









LoRaWAN longest range

height = 38.000 m

flights = 8.500 ÷ 12.000 m

Tx power = 25 dBm

WiFi Laptop = 15 dBm





Urban scenario: Padova



Need to send a few bytes occasionally and want years of battery life? $\rightarrow$ LoRa

LoRa vs. LoRaWAN

LoRa: PHY layer

LoRaWAN: MAC, NWK, and APP built on LoRa

LoRaWAN is not the only MAC/NWK implementation: Symphony Link—proprietary

LoRa/LoRaWAN basics

LoRa = PHY layer $\rightarrow$ proprietary

LoRa= Long Range $\rightarrow$ up to tens of km

Based on CSS (Chirp Spread Spectrum)

Low bit rate: $0.25 \div 11$ kbps

ISM bands: EU 868 MHz $\rightarrow$ anyone can deploy

LoRaWAN = MAC layer $\rightarrow$ open

Access method: ALOHA

Duty cycle 1% w/o channel access method

Short payload: max $59 \div 230$ bytes (in EU)

Non IP-compliant

PHY layer – LoRa

LoRa modulation

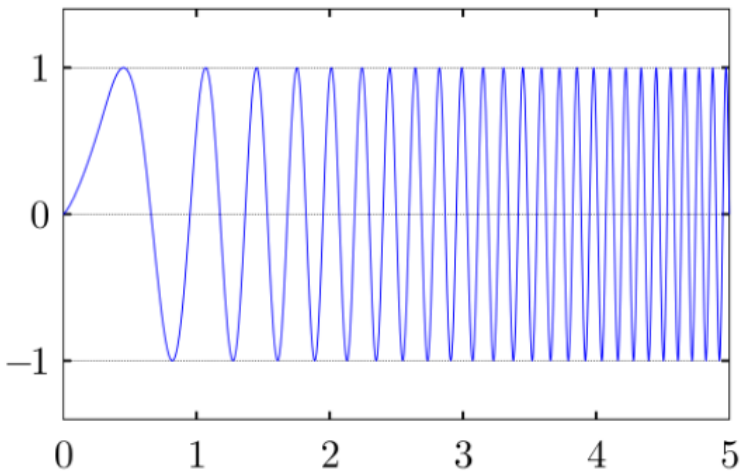
Based on Chirp Spread Spectrum (CSS)



LoRa modulation

Based on Chirp Spread Spectrum (CSS)

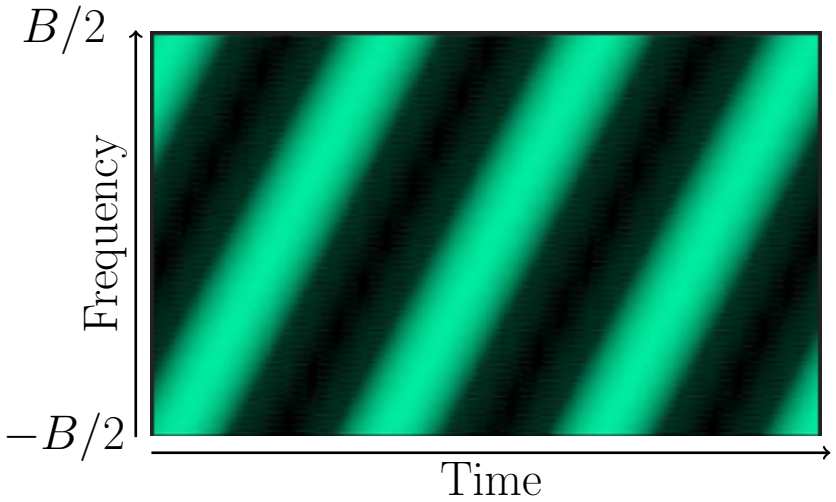




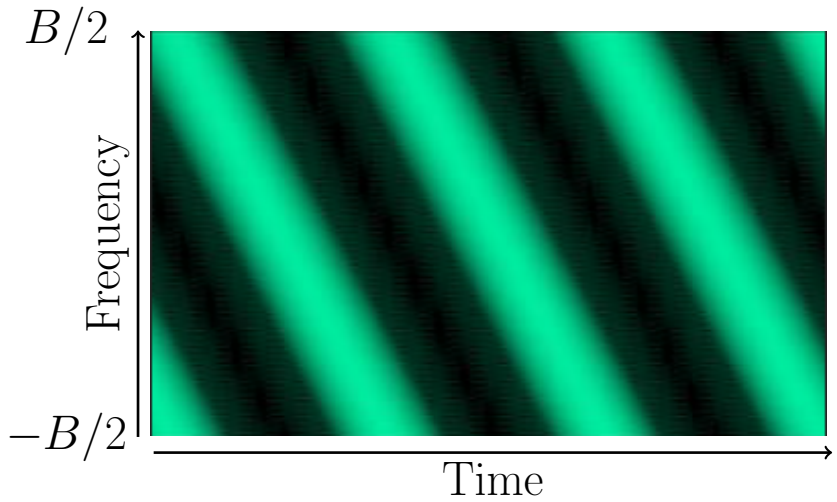
▸ Chirp sound

Up-chirp / down-chirp

Up-chirp: $e^{2\pi jt(f_0+at)}$



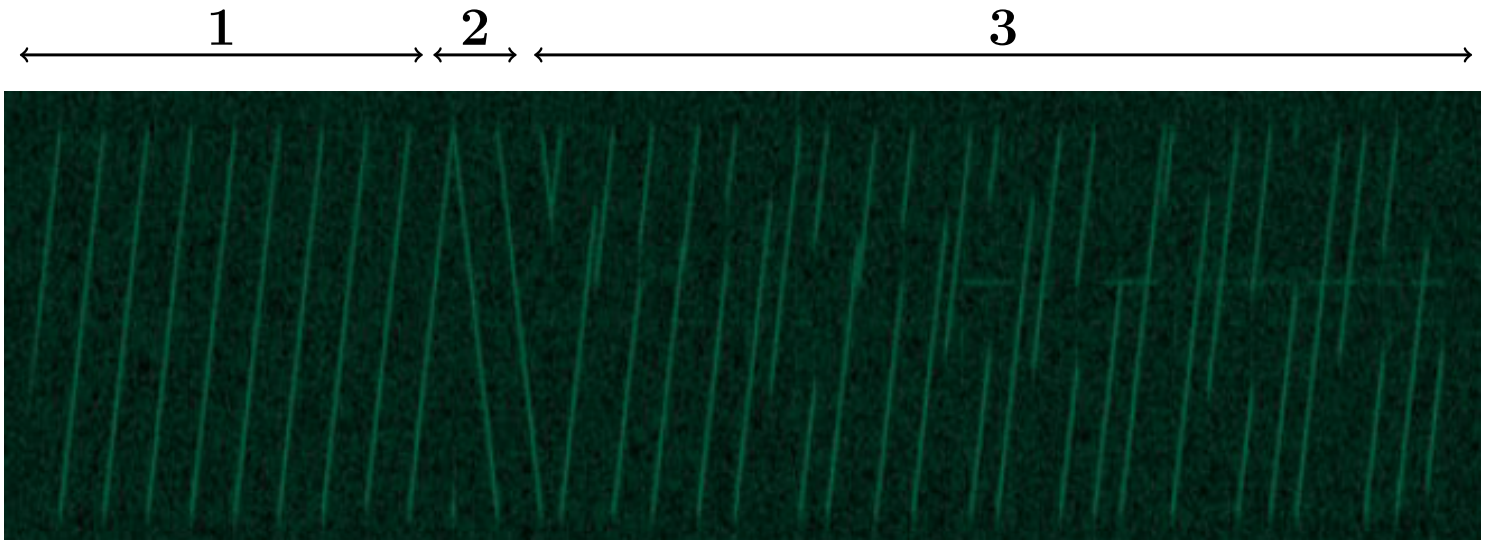
Down-chirp: $e^{-2\pi jt(f_0+at)}$



$$e^{2\pi jt(f_0+at)} \times e^{-2\pi jt(f_0+at)} = 1$$

LoRa Frame

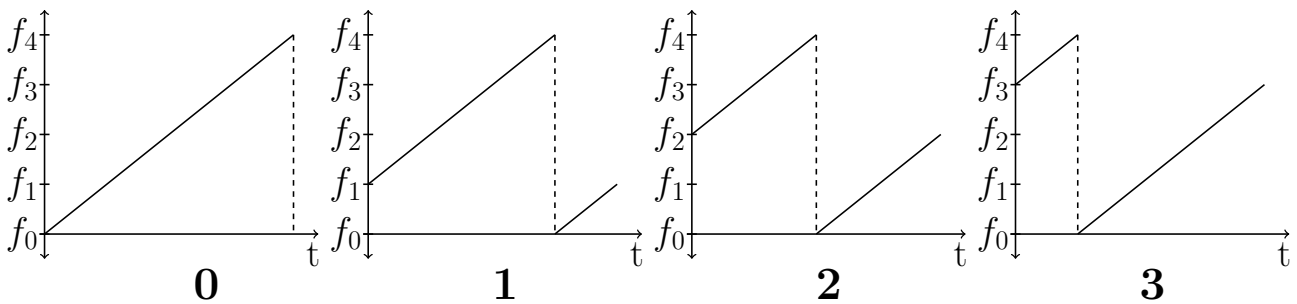
1. Preamble, training sequence
2. Beginning of the PHY data unit
3. Actual data



- The preamble is used for:
  - Time synchronization (Δt estimation)
  - Carrier frequency offset estimation (f_0)
  - Symbol rate alignment

Data encoding before transmission

- Bits are grouped into SF-bit symbols (2^{SF} possible values)
- Each symbol is mapped to a frequency shift (CSS modulation)

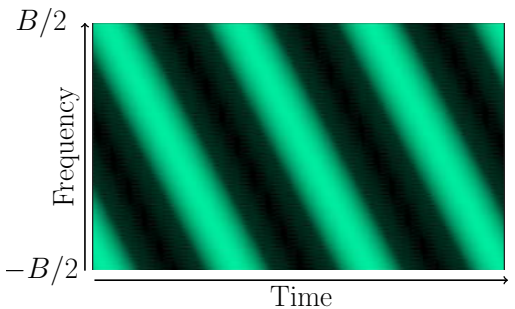


Example SF = 2, $2^{SF} = 4$ symbols

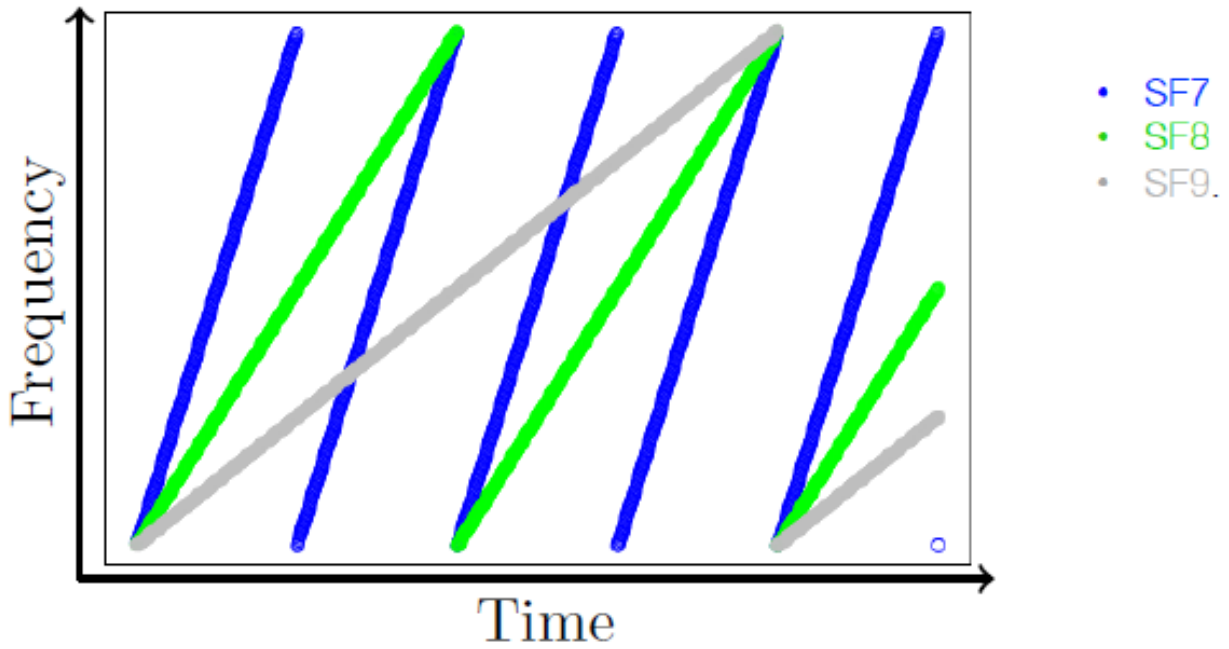
- Signal robustness techniques: Gray coding (reduce bit errors) and whitening (avoid long patterns)
- Error protection: interleaving and FEC (coding rate 4/5 to 4/8)

LoRa Demodulation

- Receiver performs **dechirping**:
  - multiply received signal by a reference down-chirp
  - convert chirps into constant tones (pure frequencies)
- FFT is then used to detect the frequency peak
- Peak position $\rightarrow$ transmitted symbol
- Key idea: **chirp correlation converts time structure into frequency**
- Enables reliable decoding at very low SNR



Spreading Factor (SF)



- SF controls how long a symbol is spread in time
- Higher SF $\rightarrow$ lower data rate, higher robustness
- Each +1 in SF roughly halves data rate and increases airtime

Spreading Factor (SF)

Key relationships

- Number of symbols: 2^{SF}
- Symbol duration: $D = \frac{2^{SF}}{B}$
- Symbol rate: $R_s = \frac{B}{2^{SF}}$
- Bit rate: $R_b = SF \cdot R_s = \frac{SF \cdot B}{2^{SF}}$



Data-rates and sensitivity

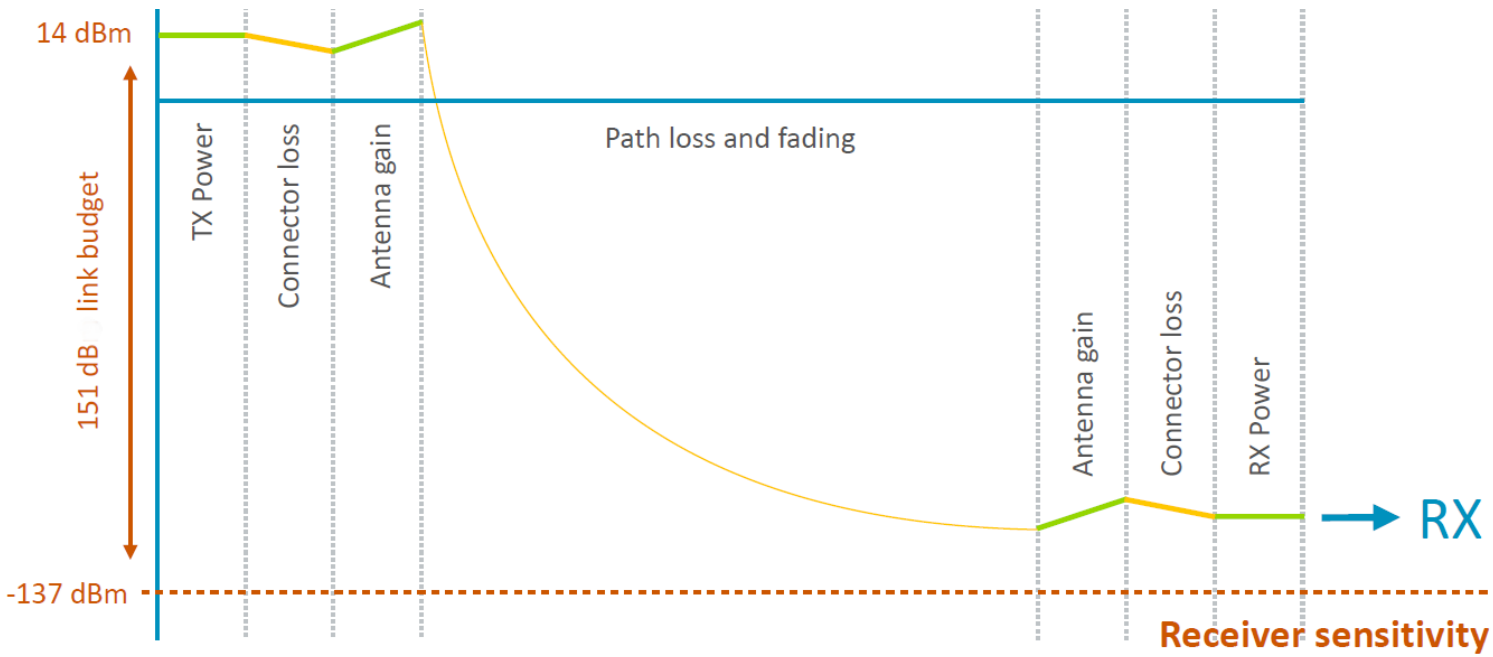
Assuming an antenna with 6 dBi reception gain

Data-rate LoRaWAN	SF	Sensitivity [dBm]
DR5	7	-134.5
DR4	8	-137.0
DR3	9	-139.5
DR2	10	-142.0
DR1	11	- 144.5
DR0	12	-147.0

Sensitivity increases of 2.5 dB with SF

Link budget comparison

	TX Power	RX Sensitivity	Link budget
WiFi	20 dBm	−75 dBm	**95 dB**
Sub-GHz 6LoWPAN	11 dBm	−110 dBm	**121 dB**
LoRa	14 dBm	−137 dBm	**151 dB**



Theoretical max distance in free space

Friis formula

$$P_r(d) = P_t G_t G_r \left(\frac{\lambda}{4\pi d}\right)^{\nu} = P_t G_t G_r \left(\frac{c}{4\pi d f_c}\right)^{\nu}$$

with $\nu = 2 \div 4$ 2 $=$, path-loss exponent.

Example with $\nu = 2 \rightarrow$ free space

Wi-Fi: 2.4 GHz, with 95 dB link budget ($P_t - P_r$)

550 meters

LoRa: 868 MHz, with 151 dB link budget ($P_t - P_r$)

870 kilometers

LoRaWAN trade-offs

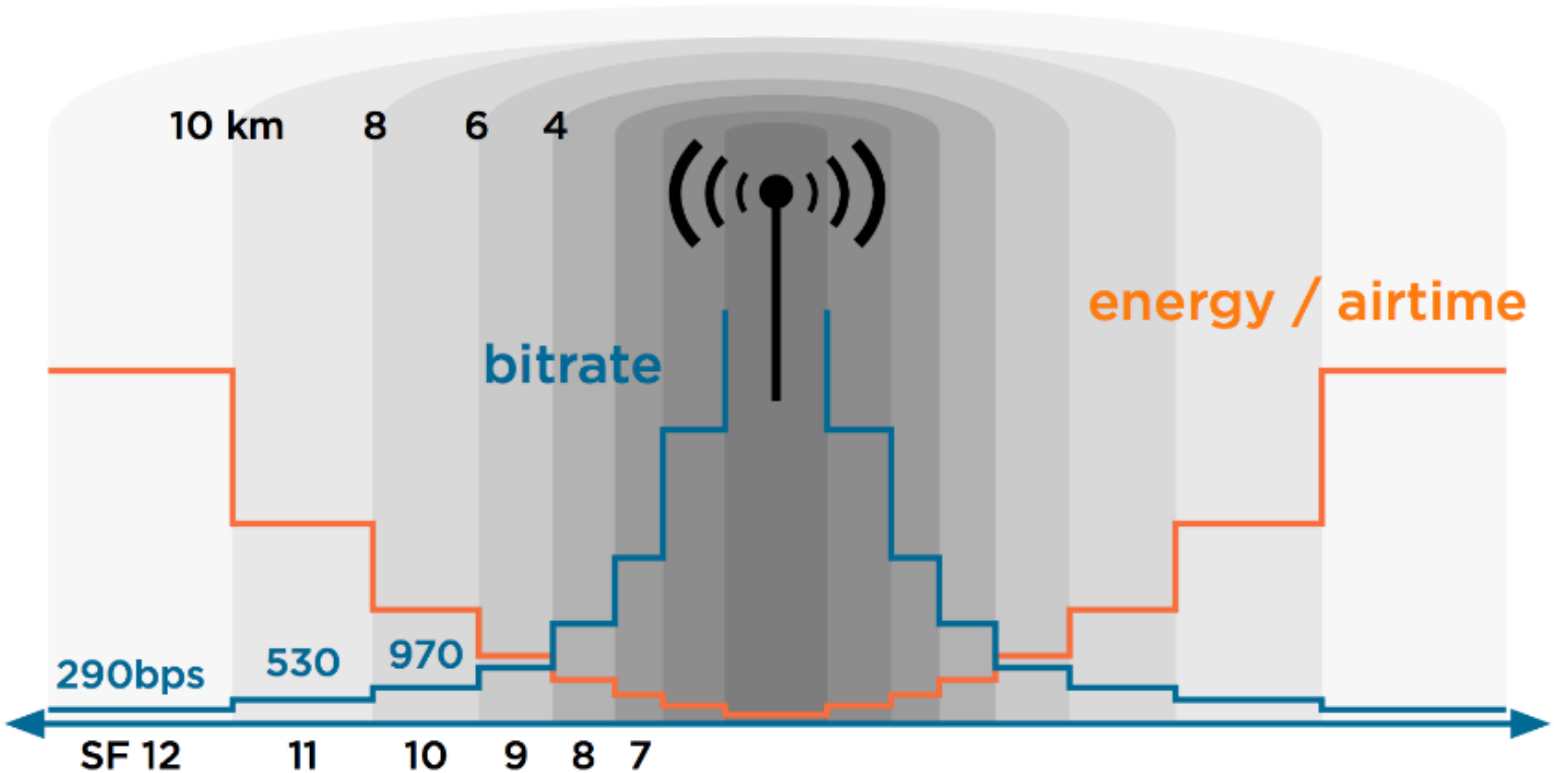
- Higher Spreading Factor $\rightarrow$ longer airtime
- Longer airtime $\rightarrow$ higher collision probability
- Higher SF $\rightarrow$ lower data rate but better sensitivity

Key trade-off

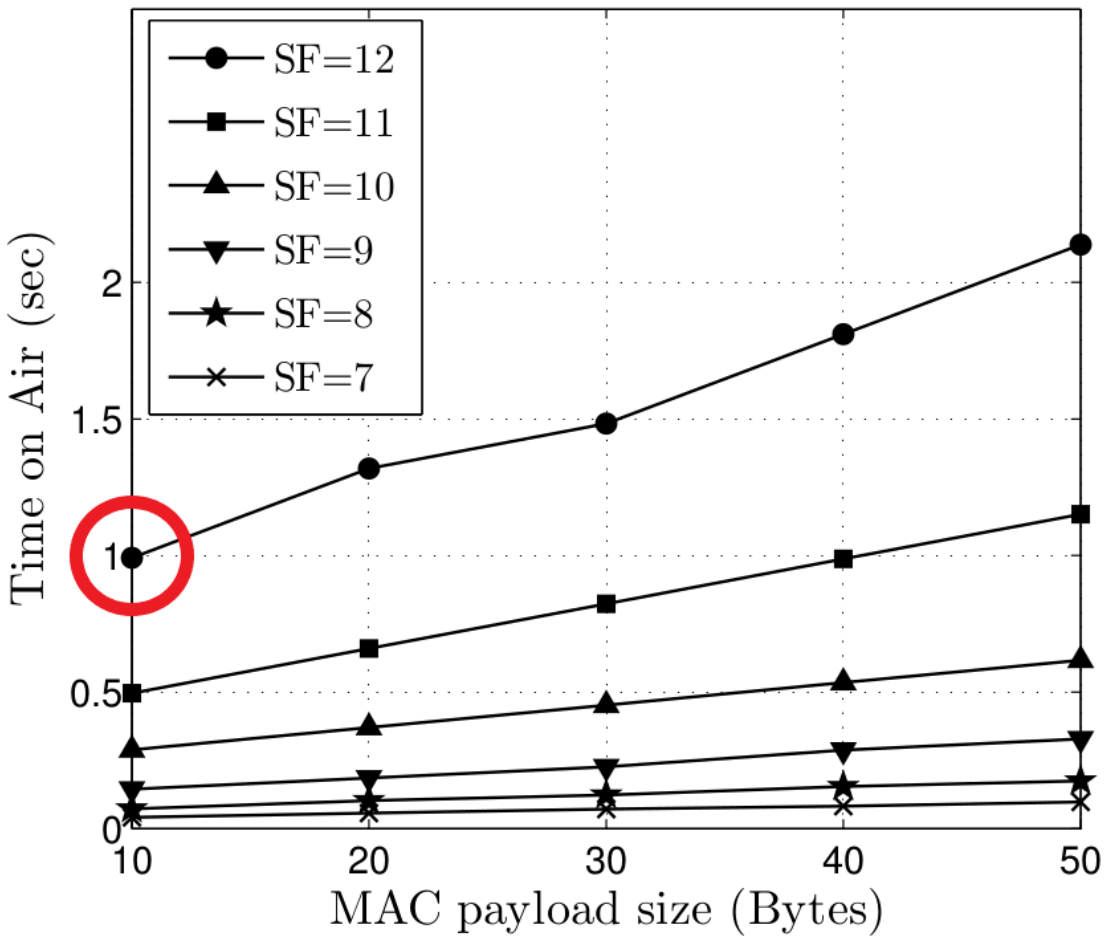
$$\text{Range} \uparrow \longleftrightarrow \text{Time-on-air} \uparrow \longleftrightarrow \text{Battery lifetime} \downarrow$$

- LoRaWAN is not about throughput, but about efficient sparse transmissions

A complete sketch



T_{air} @ code rate 4/5 and BW = 125 kHz



For US915 @ SF = 12, $PL_{\max} = 11$ bytes

In C, 1 char = 1 byte $\rightarrow$ Hello world

Data representation: JSON

Basic example: 54B

{“pressure”: 997.8,“temperature”: 26.6,“humidity”: 55.5}

Short example: 36B

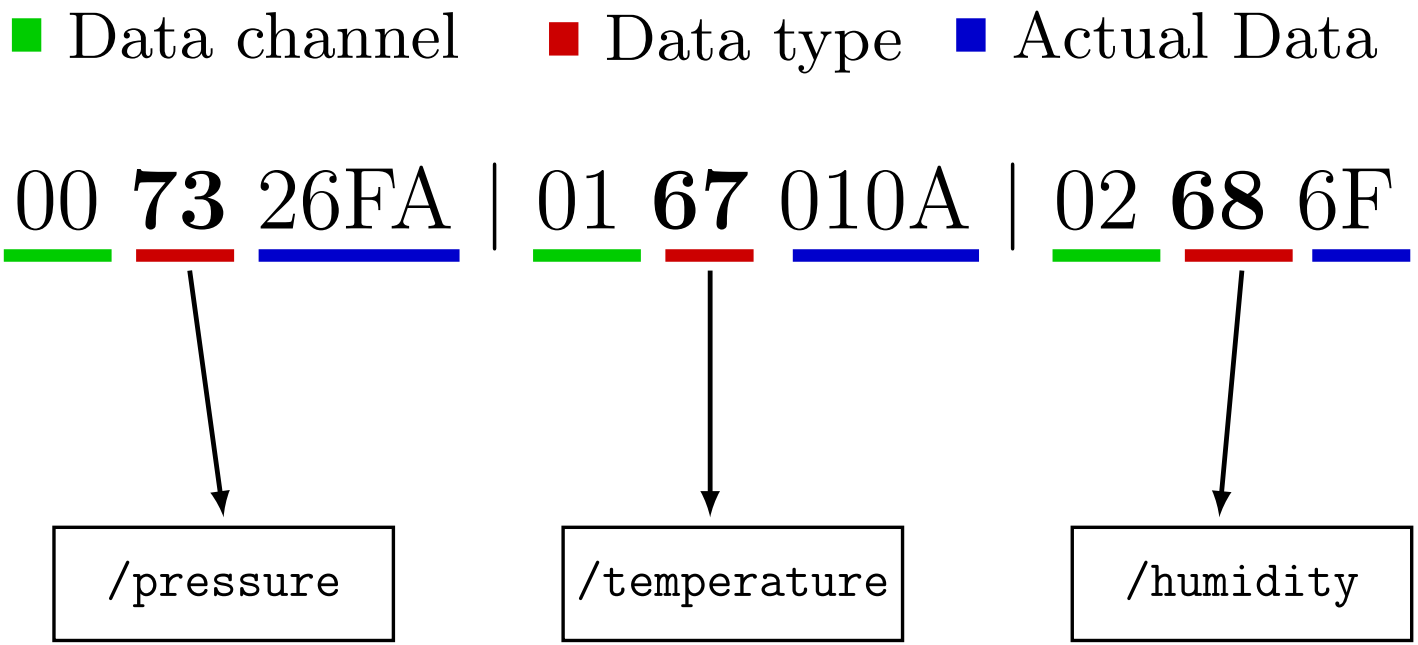
{“prs”: 997.8,“tmp”: 26.6,“hum”: 55.5}

Shortest example: 30B; one letter per sensor: limited

{“p”: 997.8,“t”: 26.6,“h”: 55.5}

Data representation: LPP

Cayenne Low Power Packet (LPP) format



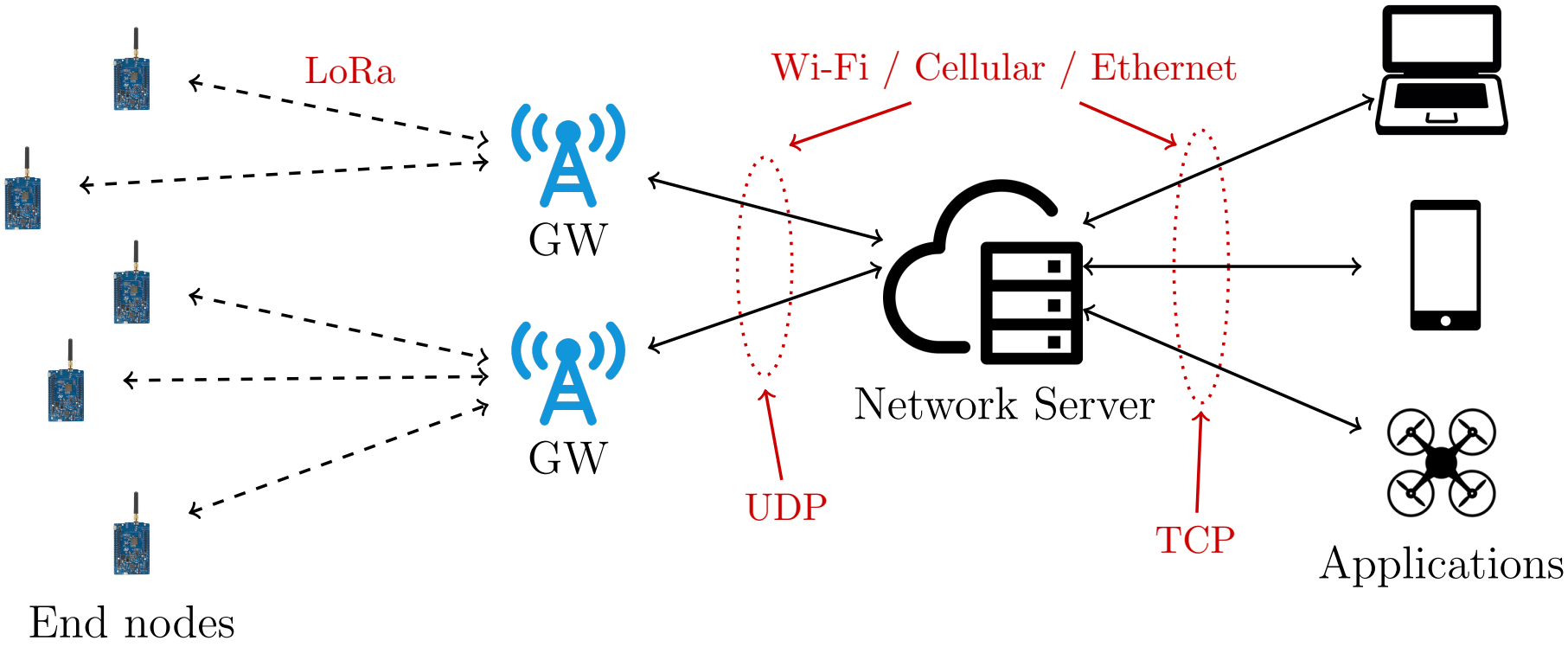
Only 11 bytes!

LPP format

Sensor	HEX value	Length [bytes]	Data Resolution per bit
Analog input	`0x02`	2	0.01 Signed
Temperature	`0x67`	2	0.1 °C Signed MSB
Humidity	`0x68`	1	0.5% Unsigned
Accelerometer	`0x71`	6	0.001 G Signed MSB per axis
Barometer	`0x73`	2	0.1 hPa Unsigned MSB
GPS	`0x88`	9	Lat: 0.0001 ° Signed MSB Lng: 0.0001 ° Signed MSB Alt: 0.01 m Signed MSB

MAC layer – LoRaWAN

Architecture: star of star



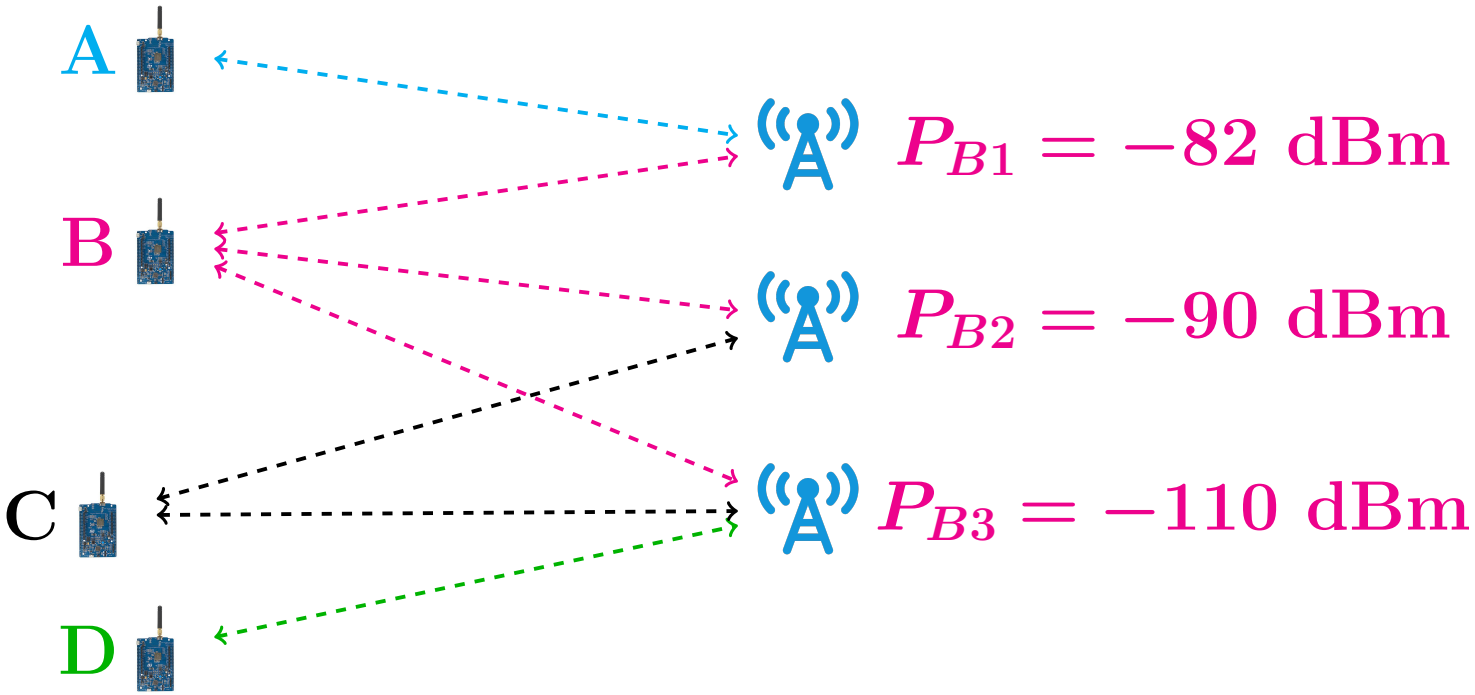
Used by...

- TTN, crowd-sourced public network
- KPN (Netherlands), Orange (France), A2A Smart City (Italy), ...
- Whoever wants to deploy it!

Remark

KPN and Orange are also mobile network operators

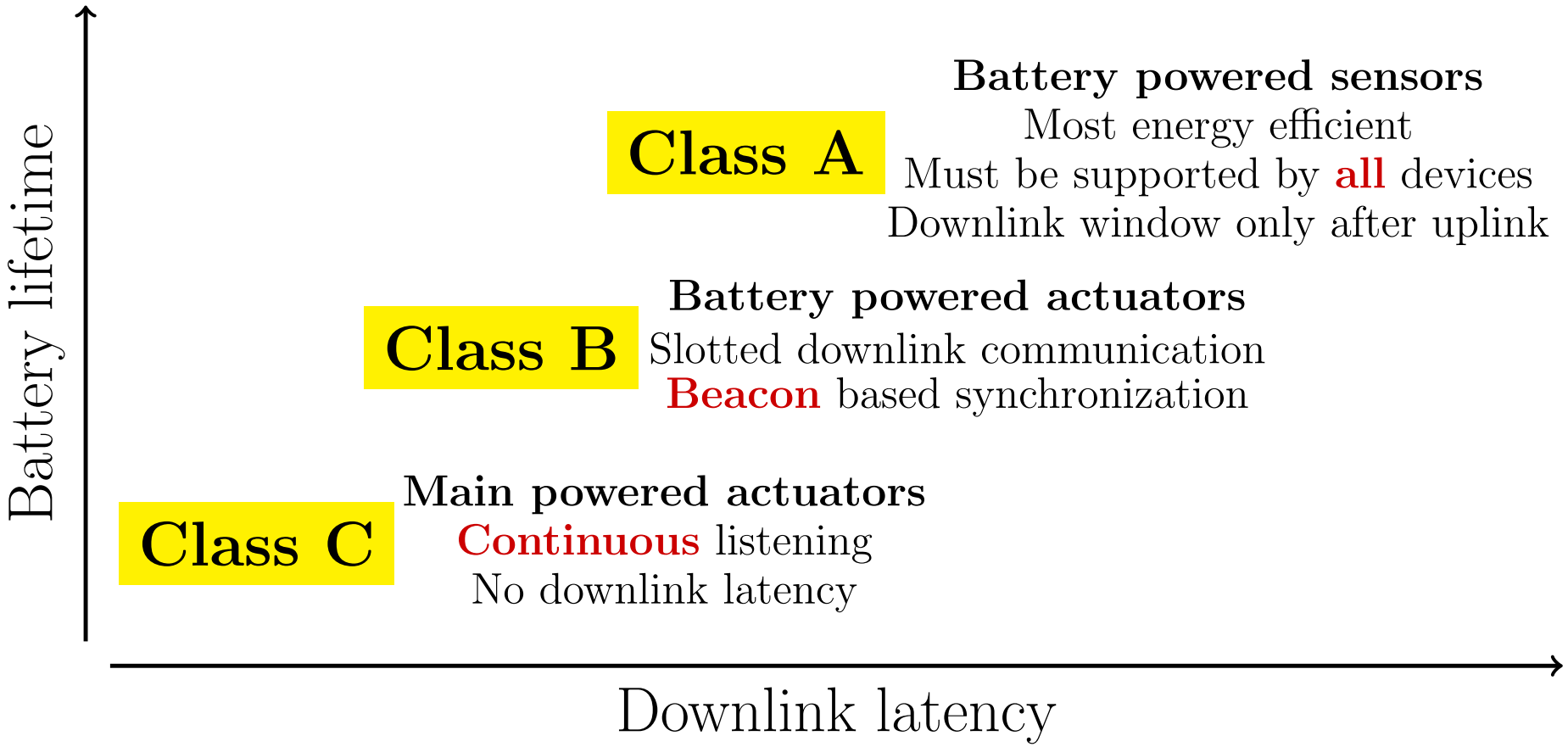
Uplink transmissions



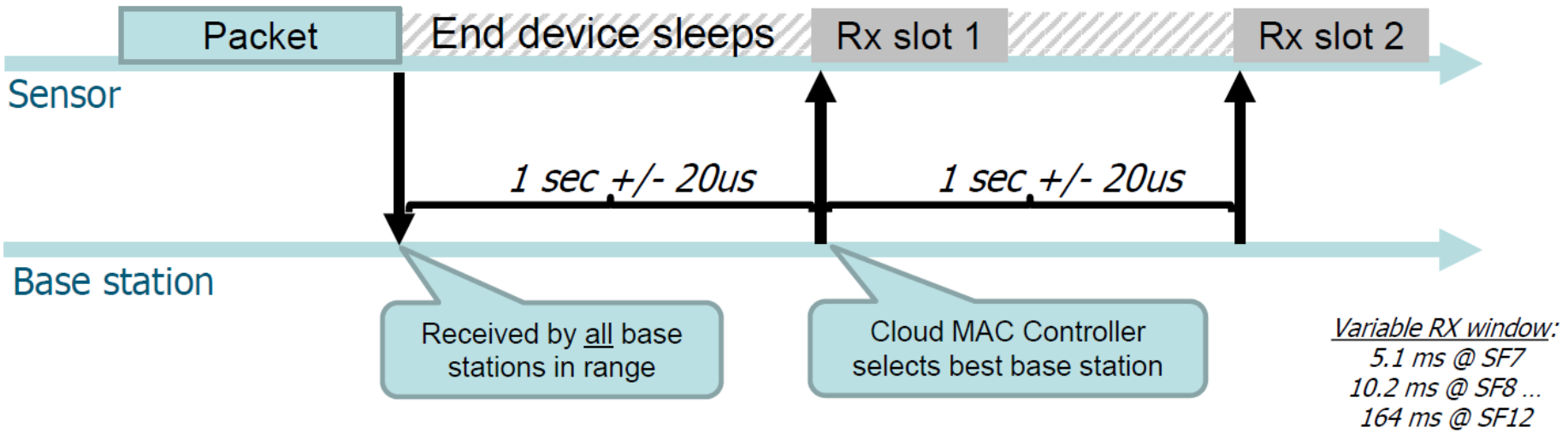
A packet is received by all the GW in the range of the node

Example: packet from B is received by all the GWs
The NS will further process it

Device classes



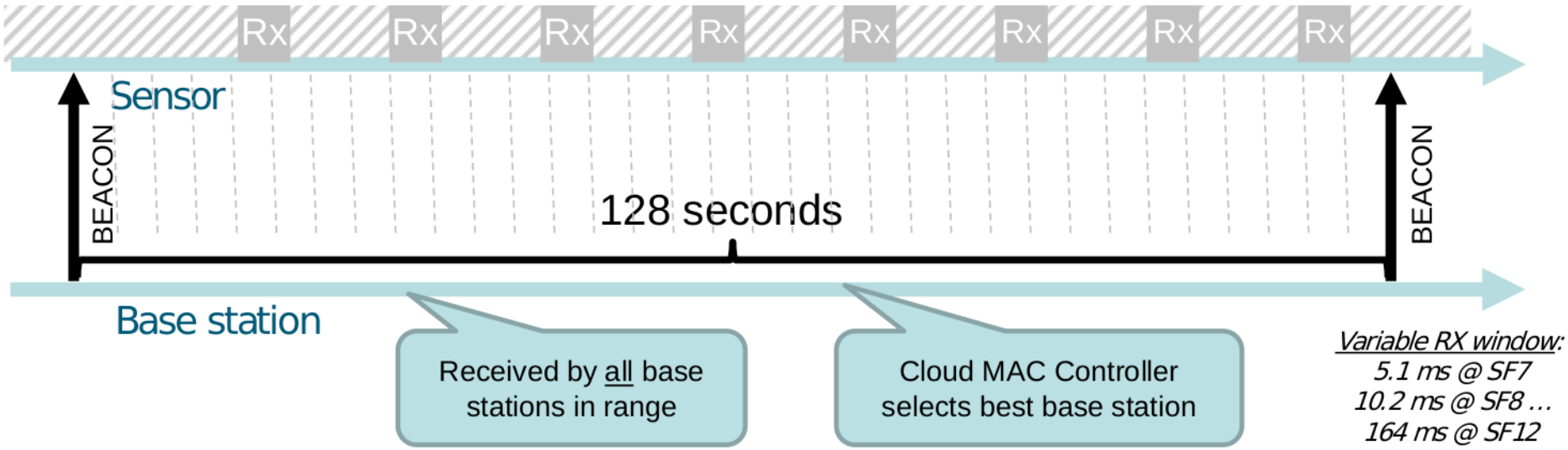
Class A devices



2 RX windows follow the transmission

- at t_1 s (by default $t_1 = 1$s) (same channel as transmission)
- at $t_1 + 1$ s (channel and SF fixed in advance)
  - by default: RX2 at 869.525 MHz (center of h1.6), DR0 (SF12, 125 kHz)

Class B devices



All sensors are **synchronized** with the gateways

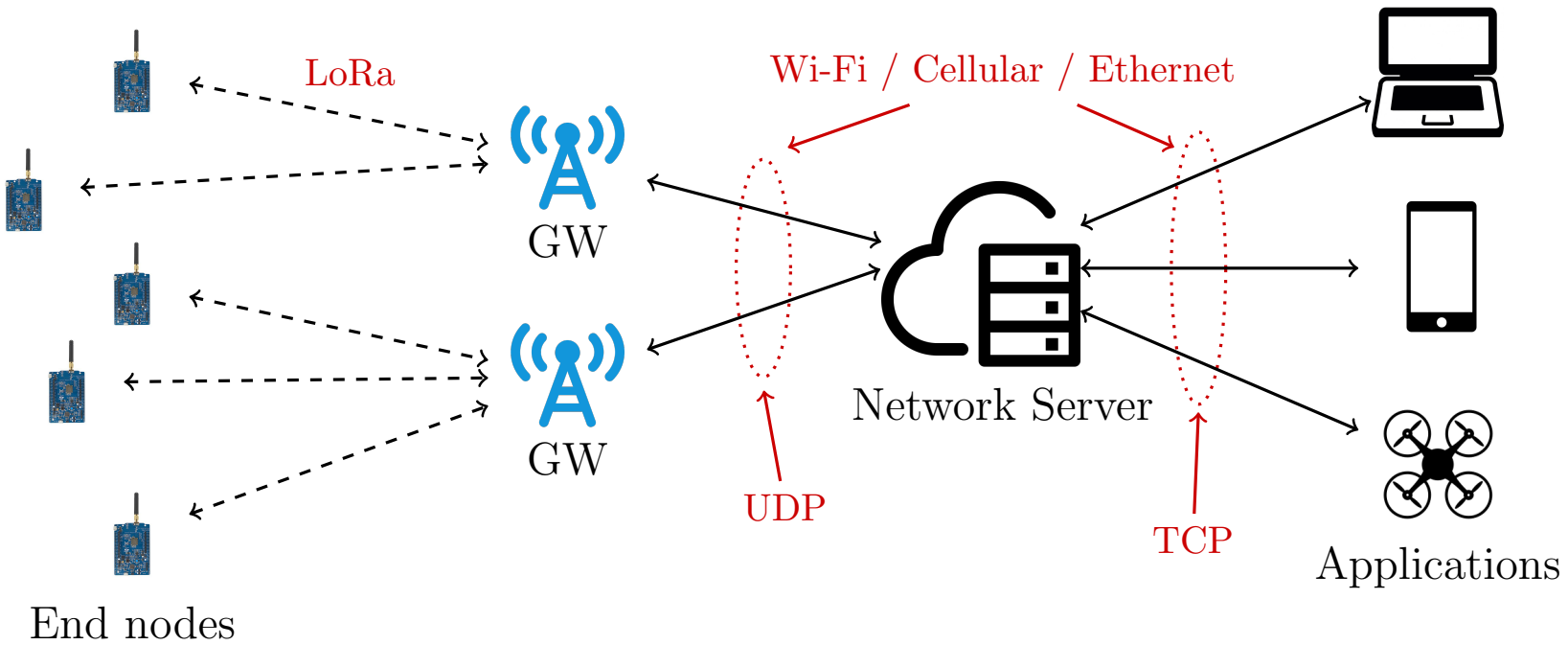
Basic working principles

- ALOHA-based access (low duty-cycle, no scheduling)
- Devices operate in a wake-up $\rightarrow$ transmit $\rightarrow$ sleep cycle
- Designed for ultra-low power operation

- Uplink can be:
  - Unconfirmed (best effort)
  - Confirmed (requires ACK from gateway)

Network Architecture and Security

Network architecture



Devices


SEMTECH
LoRa Picocell Gateway
RX
TX
ANT


pycom
GO INVENT
WiFi
Bluetooth
LoRa
Sigfox
lopy
pycom


HDMI

LoRaWAN Network: data path

- **Device:** sends uplink data and receives downlink messages
- **Gateway:**
  - Receives radio packets
  - Forwards them to the Network Server
  - Transparent (no decision logic)
- **Network Server (control core):**
  - Manages sessions and join procedure
  - Removes duplicates
  - Performs ADR (link optimization)

LoRaWAN Network: control and application

- **Network Server:**
  - Selects gateway and timing for downlink
  - Handles MAC-layer control

- **Application Server:**
  - Handles application logic
  - Decrypts payload
  - Generates responses to devices

Network Server and Application Server can be co-located or separated

Adaptive Data Rate (ADR)

- ADR is a **Network Server mechanism** to optimize link efficiency
- It balances data rate, airtime, and energy consumption

- Key parameters:
  - Spreading Factor
  - Transmission Power

Adaptive Data Rate (ADR mechanism)

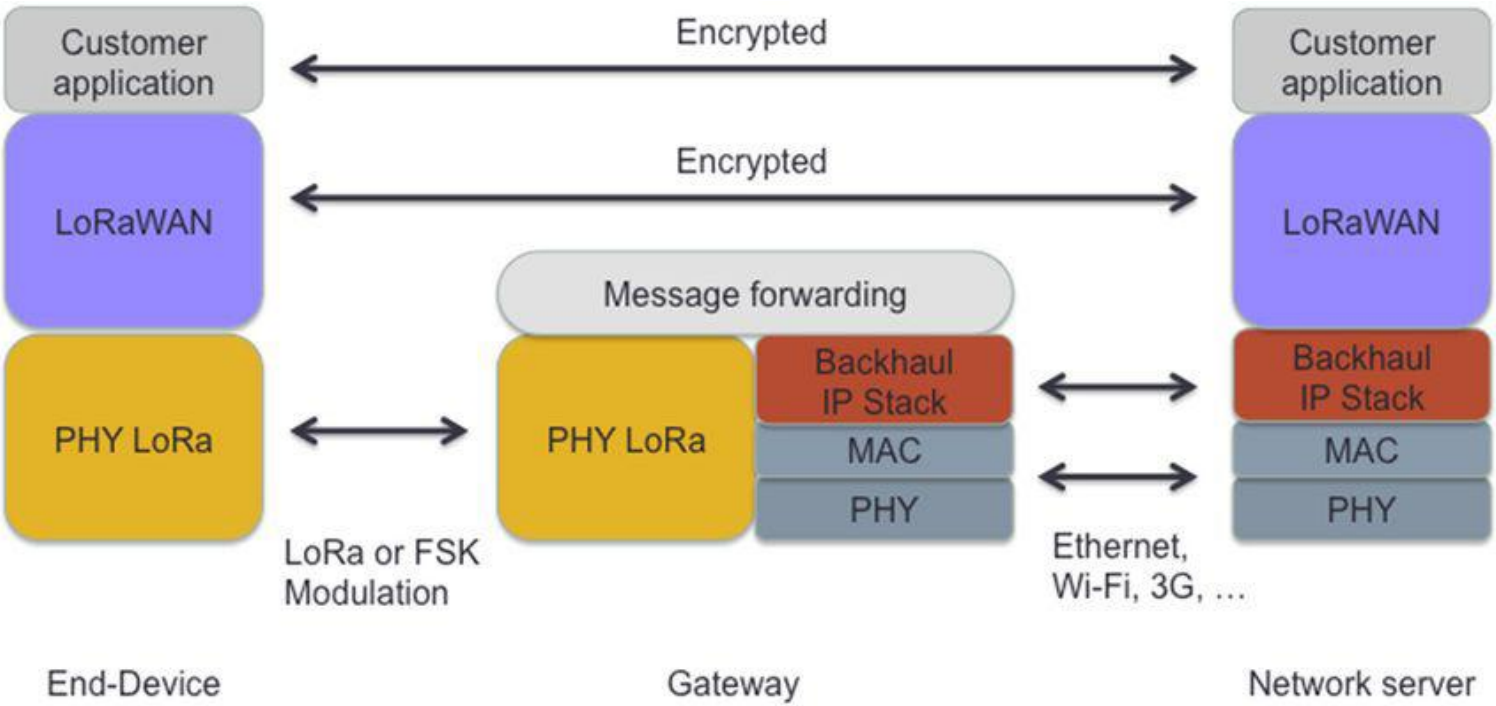
- Network Server estimates link quality (e.g. SNR over time)
- Computes optimal SF and transmission power
- Sends MAC command (LinkADRReq) to the device

- Good channel conditions $\rightarrow$ lower SF, higher data rate
- Poor channel conditions $\rightarrow$ higher SF for reliability

Important

ADR is controlled by the Network Server, but devices can enable/disable it

Protocol stack



Remarks

Available network servers: Brocaar, Gotthard (on github)

LoRaWAN Addressing model

Device identity (hardware level)

DevEUI: 64-bit unique device identifier *(like a MAC address)*

Network identity (routing level)

DevAddr: 32-bit address assigned by the Network Server *(like an IP address)*

Application identity

AppEUI: identifies the application provider (JoinEUI in newer specs)

Application routing

FPort: selects the application inside the device *(like TCP/UDP port)*

Node activation (Join procedure)

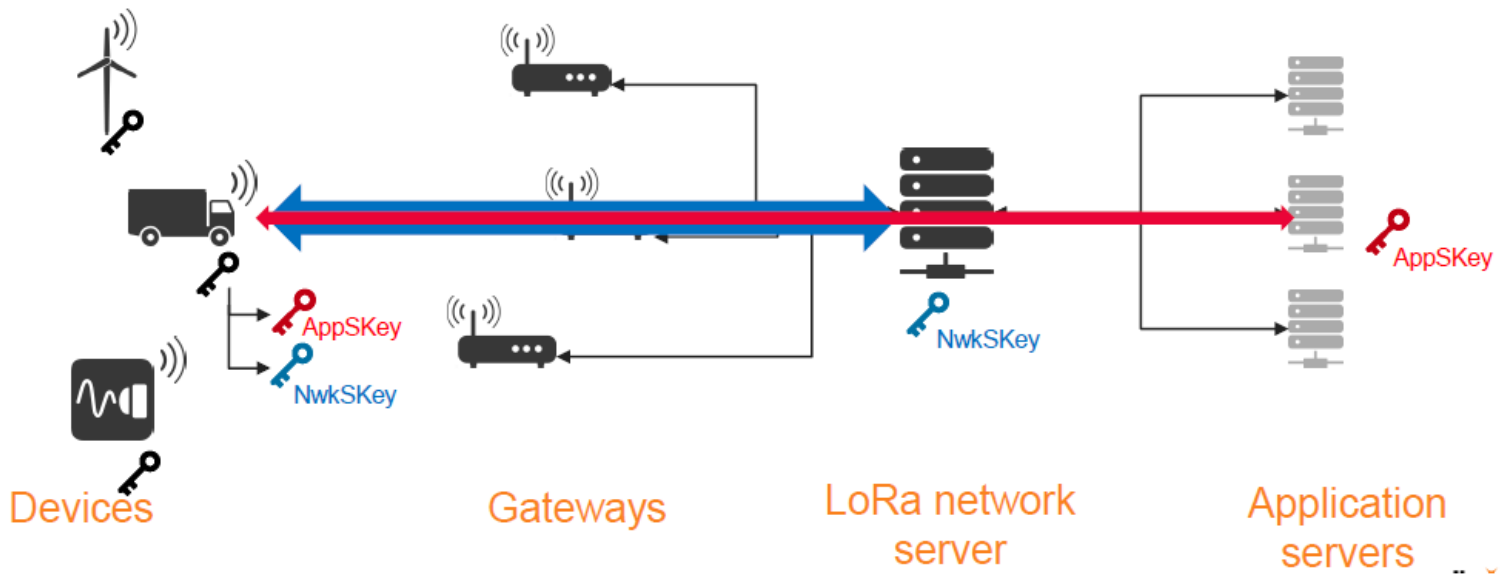
Before communicating on a LoRaWAN network, a device must be activated.

- This process establishes:
  - **DevAddr** (network address)
  - **NwkSKey** (network session key)
  - **AppSKey** (application session key)

- Two activation methods:
  - **OTAA** (Over-The-Air Activation) — dynamic and secure
  - **ABP** (Activation By Personalization) — pre-configured

Keys

- NwkSKey: shared between Device and Network Server (msg integrity)
- AppSKey: shared between Device and Application Server (end-to-end security)



OTAA (Over-The-Air Activation)

- Secure join procedure used when a device connects to the network
- Session keys are generated dynamically at each join

- Join flow:
  - Device sends **Join Request**
  - Network Server authenticates the device
  - Session keys are created
  - Device receives and decrypts **Join Accept**

OTAA: advantages and limitations

Advantages

- Strong security (fresh session keys at each join)
- Scalable key management
- Recommended for real deployments

Limitations

- Join procedure required at each (re)connection
- Slight latency during activation
- Higher implementation complexity (need server for key)

ABP (Activation By Personalization)

- No join procedure at network attachment
- Device is pre-configured offline (no handshake)

- Pre-installed parameters:
  - DevAddr
  - NwkSKey
  - AppSKey

- Keys remain fixed for the entire device lifetime unless manually changed

ABP: advantages and limitations

Advantages

- Very simple deployment
- No join delay (instant communication)
- Useful for testing or constrained environments

Limitations

- No key renewal or rotation
- Poor scalability in large deployments
- Higher security risk if keys are exposed

OTAA vs ABP: key differences

- **1. Join procedure**
  - OTAA: full network handshake (Join Request / Join Accept)
  - ABP: no handshake, direct activation
- **2. Key management**
  - OTAA: session keys generated dynamically at each join
  - ABP: keys are pre-installed and static
- **3. Device state management**
  - OTAA: Network Server maintains session state and join lifecycle
  - ABP: no session lifecycle management (static configuration)
- **4. System complexity**
  - OTAA: requires join protocol + key management infrastructure
  - ABP: minimal logic, manual provisioning

Beyond the App Server

- Rest API
- MQTT
- Web-socket

Platforms

- https://www.chirpstack.io/
- https://www.thethingsnetwork.org/

Current and Future works

Scalability: From Pilot to Production

- $>$125 million devices deployed globally
- $\sim$25% CAGR (compound annual growth)
- Multi-million device deployments in:
  - Utilities (smart metering)
  - Smart cities
  - Industrial IoT
- **Key point:** LoRaWAN has moved beyond pilot phase

LoRaWAN over Satellite (NTN)

- Emerging integration with **Non-Terrestrial Networks (NTN)**
- Currently satellite used as IP backbone
- Direct device-to-satellite communication (GW on the satellite, experimental)

- Enables:
  - Global coverage (remote areas, oceans, deserts)
  - Infrastructure-free deployments

Limitations

- Very constrained link budget
- Limited downlink capacity
- Non-deterministic connectivity (orbital constraints)

Protocol Evolution

Data Rate and Capacity

- Introduction of multi-band operation:
  - sub-GHz $\rightarrow$ long range
  - 2.4 GHz $\rightarrow$ higher throughput
- LR-FHSS (frequency hopping) for improved scalability

Security and Interoperability

- Improved certification processes
- Updated security specifications (e.g., LoRa Alliance TR007)
- Focus on reducing ecosystem fragmentation

Question for you:
What is the most relevant topic right now?

From sensing to decisions

- LoRaWAN acts as a **data acquisition layer**
- Devices remain simple: sense $\rightarrow$ transmit
- Intelligence is centralized in backend systems

- Typical architecture:

  Sensors $\rightarrow$ LoRaWAN $\rightarrow$ Cloud / Analytics / AI $\rightarrow$ Decisions

- Enables:
  - Predictive maintenance
  - Smart monitoring systems
  - Infrastructure optimization

Industrial Integration

- Convergence with industrial standards
- Mapping toward OPC UA (industrial communication standard)
- Integration into OT/SCADA environments
- **Impact:**
  - Increased industrial adoption
  - Better interoperability with existing systems

Network Architecture Evolution

Hybrid Networks

- Combination of:
  - Public networks
  - Private deployments
  - Satellite connectivity

Backhaul Innovation

- Use of Wi-Fi HaLow for long-range backhaul
- Reduced infrastructure costs

Mesh Research

- Multi-hop/mesh under study
- Not part of current standard

Market Growth

- Market size:
  - $\sim$ \$6.7B (2026)
  - $\sim$ \$54B (2033)
- Strong expansion across industries
- **This means:**
  - No longer an experimental technology
  - Established IoT connectivity solution

Critical Summary

Real Advances

- Large-scale deployment
- Satellite integration
- Multi-band support
- Industrial ecosystem integration

Limitations

- Not high-throughput
- No standardized mesh
- Limited latency and QoS

Conclusion: LoRaWAN is evolving, but remains an LPWAN technology

If you want to know more

- **Curiosity is the best protocol**
- https://www.youtube.com/watch?v=NoquBA7IMNc
- Understanding the Limits of LoRaWAN, by Ferran Adelantado
- https://developers.mydevices.com/cayenne/docs/lora/#lora-cayenne-low-power-payload
- https://www.thethingsnetwork.org/docs/lorawan/

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Overview on LoRa/LoRaWAN

